



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
04-04-2021

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, LIVE-I(A) & LIVE-I]

PAPER-1

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

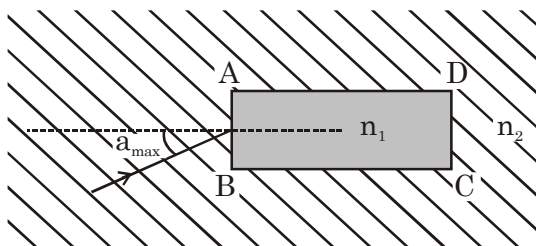
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. A rectangular glass slab ABCD, of refractive index n_1 , is immersed in water of refractive index n_2 ($n_1 > n_2$). A ray of light is incident at the surface AB of the slab as shown. The maximum value of the angle of incidence $a_{\max}$, such that the ray comes out only from the other surface CD is given by



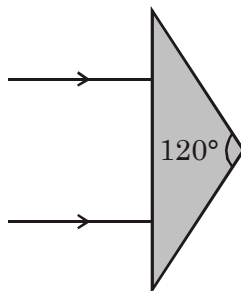
- (A) $\sin^{-1} \left[\frac{n_1}{n_2} \cos \left(\sin^{-1} \frac{n_2}{n_1} \right) \right]$ (B) $\sin^{-1} \left[n_1 \cos \left(\sin^{-1} \frac{1}{n_2} \right) \right]$
- (C) $\sin^{-1} \left(\frac{n_1}{n_2} \right)$ (D) $\sin^{-1} \left(\frac{n_2}{n_1} \right)$

Space for Rough Work

2. Two particles are performing SHM about same mean position and have same amplitude A. At $t = 0$ both particle have same velocity and separation between them is $\sqrt{3}A$ then at which time they cross each other if time period of SHM for both particle is T.
- (A) $\frac{T}{4}$ (B) $\frac{T}{8}$ (C) $\frac{T}{12}$ (D) $\frac{T}{16}$
3. A particle is performing vertical circular motion with a time period T. Then time taken by particle to travel lower half of the circle is :
- (A) Equal to $\frac{T}{2}$ (B) Greater than $\frac{T}{2}$ (C) Less than $\frac{T}{2}$ (D) Can't say
4. A convex lens of focal length f is placed somewhere in between an object and a screen. The distance between the object and the screen is x. If the numerical value of the magnification produced by the lens is m, then the focal length of the lens is :
- (A) $\frac{mx}{(m+1)^2}$ (B) $\frac{mx}{1-m^2}$ (C) $\frac{(m+1)^2}{m}x$ (D) $\frac{(m-1)^2}{m}x$

Space for Rough Work

5. A person is in a room whose ceiling and two adjacent walls are mirrors. How many images are formed
 (A) 5 (B) 6 (C) 7 (D) 8
6. An isosceles prism of angle 120° has a refractive index of 1.44. Two parallel monochromatic rays enter the prism parallel to each other in air as shown. The rays emerging from the opposite faces :



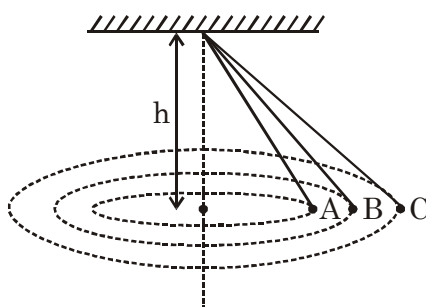
- (A) Are parallel to each other
 (B) Are diverging
 (C) Make an angle $2\sin^{-1}(0.72)$ with each other
 (D) Make an angle $2\{\sin^{-1}(0.72) - 30^\circ\}$ with each other

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. All three point masses A, B and C shown in the figure have equal mass and performing circular motion in concentric circles shown in figure. If T_A , T_B and T_C represent the tension in strings which are connected to point mass A, B and C respectively and ω_A , ω_B and ω_C are angular velocities then choose the **CORRECT** option(s) :



- (A) $T_A < T_B < T_C$ (B) $T_A = T_B = T_C$ (C) $\omega_A < \omega_B < \omega_C$ (D) $\omega_A = \omega_B = \omega_C$

Space for Rough Work

8. If position, velocity and acceleration of a particles are given in options as a function of time in which x_0 , v_0 and a_0 are constants and have dimensions accordingly. x , v and a represent the position, velocity and acceleration. Which of the following option(s) represent the SHM.

- (A) $x = x_0 \sin \pi t$ (B) $v = v_0 \sin(\pi t)$
(C) $a = a_0 \sin \pi t$ (D) None

9. Two different particles of same mass are performing SHM and their positrons are given as $x_1 = A \sin(\omega t + \phi)$ and $x_2 = 2A \sin(2\omega t + \phi)$ then choose the correct statement(s) :

$v_{\max} \rightarrow$ maximum speed,

$a_{\max} \rightarrow$ maximum acceleration,

$kE_{\max} \rightarrow$ maximum kinetic energy,

$PE_{\max} \rightarrow$ maximum potential energy

(A) $(v_{\max})_1 = \frac{1}{4}(v_{\max})_2$

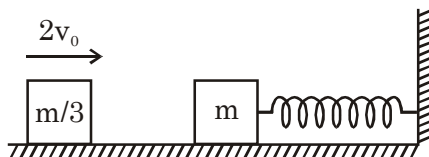
(B) $(a_{\max})_1 = \frac{1}{8}(a_{\max})_2$

(C) $(kE_{\max})_1 = \frac{1}{16}(kE_{\max})_2$

(D) $(PE_{\max})_1 < \frac{1}{8}(PE_{\max})_2$

Space for Rough Work

10. A block of mass $\frac{m}{3}$ moving with velocity $2v_0$ collides elastically to the block of mass m which is at rest and attached to spring of spring constant k . After collision block of mass m starts performing SHM then choose the **CORRECT** option(s) :
(Assume surface to be frictionless)



- (A) Maximum speed of m is v_0
- (B) Maximum acceleration of m is $v_0 \sqrt{\frac{k}{m}}$
- (C) Amplitude of SHM is $v_0 \sqrt{\frac{m}{k}}$
- (D) Time period of SHM is $2\pi \sqrt{\frac{m}{k}}$
11. An object is moving with constant speed v_0 towards a stationary concave mirror on the principle axis from far away from the pole of the mirror.
- (A) Speed of image is constant throughout the motion and equal to v_0 .
- (B) Speed of image is less than v_0 when object is in between ∞ and center of curvature.
- (C) Speed of image is greater than v_0 when object is in between center of curvature and focus.
- (D) Can't say about speed of image.
12. An equilateral prism deviates a ray through 40° for two angles of incidence differing by 10° . The angles of incidence are :
- (A) 45° (B) 55° (C) 65° (D) 70°

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

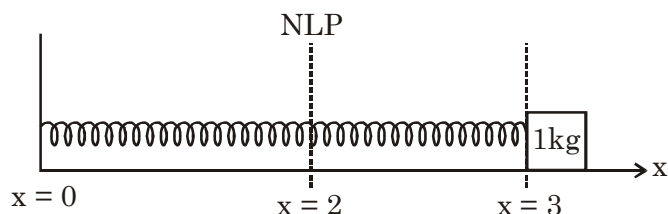
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●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
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1. A particle start performing circular motion in a circle of radius 1 m with a constant angular acceleration 2 rad/s^2 , if acceleration of particle at $t = 1 \text{ s}$ is 'a' then fill the value of a^2 in OMR.
2. A uniform rod of length 6m is hanged vertically from one end and given a small angular displacement from mean position then it starts performing angular SHM calculate the time period of that SHM. (Take : $g = \pi^2$)
3. If a particle of mass 2 kg is trapped is in a system which has potential U as a function of position x as $U = (x^2 - 4)J$ and total energy of particle is 4J then maximum possible speed of particle is : (Assume only conservative forces related to u are acting on particle)

Space for Rough Work

4. If a block of mass 1 kg is attached with a spring of natural length 2m and stiffness 100 N/m. Now block is displaced 1m in the direction of positive x-axis and released at $t = 0$ then position of block is given by $x = x_0 + A \sin\left(\omega t + \frac{\pi}{2}\right)$ then value of $\frac{x_0 + A}{\omega}$ in S.I. unit is



5. An object of size 7.5 cm is placed in front of a convex mirror of radius of curvature 25 cm at a distance of 40 cm. The size of the image (in cm) is :
6. A square of side 3 cm is placed at a distance of 25 cm from a concave mirror of focal length 10cm. The centre of the square is at the axis of the mirror and the plane is normal to the axis. The area enclosed by the image (in cm^2) of the square is :

Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 18)

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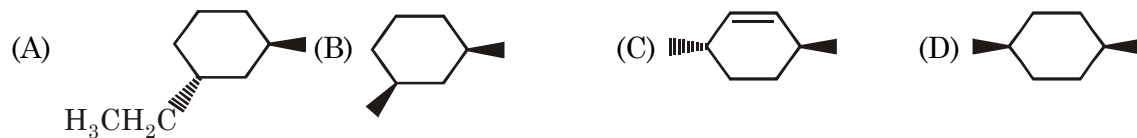
- Which of the following planar and polar molecule give atleast one oxyacid as a product on hydrolysis.
 (A) XeF_2 (B) SiCl_4 (C) ICl (D) NF_3
- The least stable resonating structure of diazomethane (CH_2N_2) is -
 (A) $\overset{\ominus}{\text{CH}_2}-\overset{\oplus}{\text{N}}\equiv\text{N}$ (B) $\overset{\oplus}{\text{CH}_2}-\text{N}=\overset{\ominus}{\text{N}}$ (C) $\overset{\ominus}{\text{CH}_2}-\text{N}=\overset{\oplus}{\text{N}}$ (D) $\overset{\oplus}{\text{CH}_2}=\text{N}=\overset{\ominus}{\text{N}}$
- A compound AB_2 crystallizes similarly as CaF_2 . $r_{\text{A}^{+2}} = 'x'$ pm $r_{\text{B}^-} = 'y'$ pm where $r_{\text{A}^{+2}}$ and r_{B^-} are respective radii. Maximum & minimum distances between two B^- ions in a unit cell are
 (A) $[(2x+2y); \frac{\sqrt{3}}{2}(x+y)]$ (B) $[(x+y); \frac{2}{\sqrt{3}}(x+y)]$
 (C) $[\frac{2}{\sqrt{3}}(x+y); (x+y)]$ (D) $[2(x+y); \frac{2}{\sqrt{3}}(x+y)]$

Space for Rough Work

4. Which of the following does not exist at room temperature?

- (A) PbCl_4 (B) PbI_4 (C) PbF_4 (D) PbI_2

5. Which of the following compound is achiral but shows optical isomerism.



6. A solution contains 31 gm of ethylene glycol in 200 gm of water. If this solution is cooled to -5.4°C . what mass of ice will separate out at this temperature ?

$[\text{K}_f \text{ for water} = 1.8 \text{ K}\cdot\text{kgmol}^{-1}]$

- (A) 33.3 g (B) 37.1 g (C) 48.2 g (D) 62.1 g

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

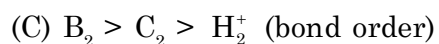
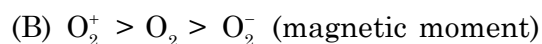
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7. Which of the following is/are planar w.r.t. underlined atom?
 (A) $\underline{\text{N}}(\text{SiH}_3)_3$ (B) $\underline{\text{Cl}}\text{O}_2$ (C) $\underline{\text{I}}_2\text{Cl}_6$ (D) $\underline{\text{Be}}_2\text{H}_4$
8. Which of the following is / are incorrect IUPAC names
 (A) 2-Ethyl-3-methyl pentane
 (B) 1-amino butane-1-one
 (C) 3-Methyl pentane
 (D) 4-(1, 1-dimethyl ethyl)-2-methyl pentane
9. For which of the following vant' Hoff's factor is correctly matched -

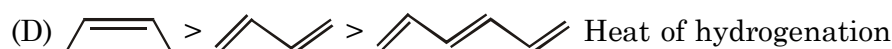
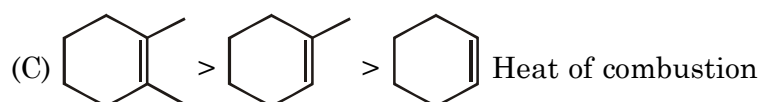
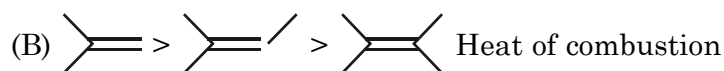
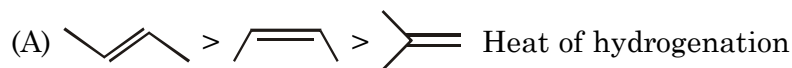
Salt	Degree of dissociation (α)	i
(A) CaCl_2	50 %	2
(B) $\text{K}_4[\text{Fe}(\text{CN})_6]$	75%	4
(C) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$	60 %	1.6
(D) KCl	90 %	2.8

Space for Rough Work

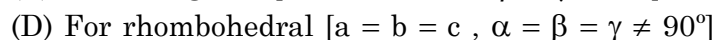
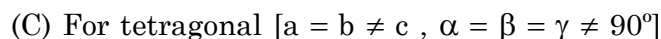
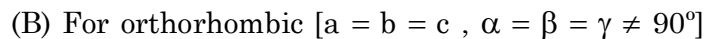
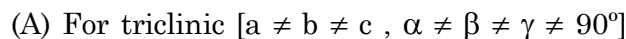
10. Select **INCORRECT** order.



11. Which of the following is **INCORRECT** :



12. Select the correct combination(s)



Space for Rough Work

SECTION-II : (Maximum Marks: 24)

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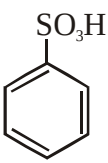
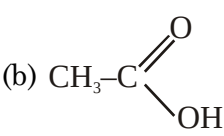
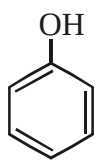
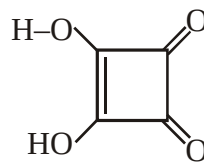
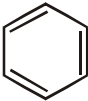
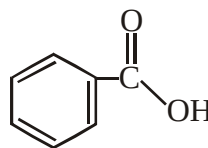
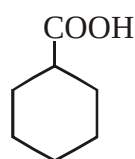
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+					-				
●	●	○	○	○	●	●	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	●
③	③	③	③	○	③	③	③	③	○
④	④	④	④	○	④	④	④	④	○
⑤	⑤	⑤	⑤	○	⑤	⑤	⑤	●	○
⑥	⑥	⑥	⑥	○	⑥	⑥	⑥	○	○
⑦	⑦	●	●	○	⑦	⑦	⑦	○	○
⑧	⑧	⑧	⑧	○	⑧	⑧	⑧	○	○
⑨	⑨	⑨	⑨	○	⑨	⑨	⑨	○	○

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If bond order for CO = x
 Total number of ABMO e^- s in O_2 = y
 and number of unpaired electrons in CN^- = z

then calculate $\frac{y+z}{x}$

2. (a)  (b)  (c)  (d) 
- (e) HCl (f)  (g)  (h) 

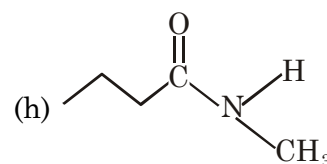
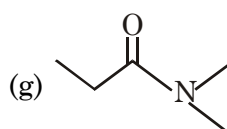
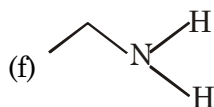
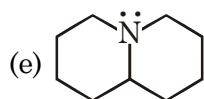
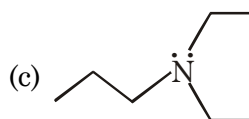
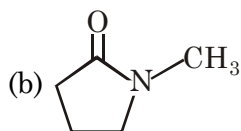
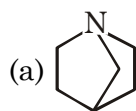
For the above compound identify the number $\alpha = x - y$

x = Number of compounds which release CO_2 on reaction with $NaHCO_3$

y = Number of compounds which are aromatic

Space for Rough Work

3. A 1 litre solution containing equal moles of FeO and $\text{Fe}_{0.8}\text{O}$ was completely titrated with 35 ml 0.3M KMnO_4 in acidic medium. Millimoles Fe^{3+} produced are
4. Find the total number of compounds which are thermally less stable than BeCO_3 .
 Li_2CO_3 , MgCO_3 , CaCO_3 , SrCO_3 , Na_2CO_3 , Rb_2CO_3 , Cs_2CO_3 , BaCO_3
5. Number of compounds with 3° amines in the following :



6. At 300K, vapour pressure of pure toluene & pure benzene are 40 and 120 torr respectively. For a solution which boils at 300K and 50 torr, % mole of benzene in solution is-

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. If the 3rd and 7th terms of an H.P. are $\frac{1}{6}$ and $\frac{1}{14}$ respectively then value of its 60th term equals
 (A) $\frac{1}{101}$ (B) $\frac{1}{120}$ (C) $\frac{1}{201}$ (D) $\frac{1}{204}$

2. Let $f : \mathbb{R} \rightarrow [-1, 1]$ and $g : \mathbb{R} \rightarrow \mathbb{B}$ and $g(x) = \sin^{-1}\left(\frac{f(x)}{2}\sqrt{4 - (f(x))^2}\right) + \frac{\pi}{3}$. If $y = f(x)$ and $y = g(x)$ are both surjective functions then set B is given by
 (A) $\left[0, \frac{\pi}{3}\right]$ (B) $\left[-\frac{\pi}{3}, \frac{\pi}{3}\right]$ (C) $[0, \pi]$ (D) $\left[0, \frac{2\pi}{3}\right]$

3. If α, β are the distinct roots of the equation $ax^2 + bx + c = 0$, ($ac \neq 0$) then $\lim_{x \rightarrow \frac{1}{\alpha}} \frac{1 - \cos(cx^2 + bx + a)}{(\alpha x - 1)^2}$
 (A) $\frac{c^2(\alpha - \beta)^2}{2\alpha^4\beta^2}$ (B) $\frac{c^2(\alpha - \beta)^2}{2\alpha^2\beta^2}$ (C) $\frac{c^2(\alpha - \beta)^2}{2\alpha^2\beta^4}$ (D) $\frac{c^2(\alpha - \beta)^2}{2\alpha^4\beta^4}$

Space for Rough Work

4. Range of the function $f(x) = \cos x \cdot \frac{\sin 4x}{\sin 4x}$ is

(A) $(-1, 1) - \left\{ -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\}$

(B) $[-1, 1] - \left\{ -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\}$

(C) $(-1, 1) - \left\{ -\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right\}$

(D) $[-1, 1] - \left\{ -\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right\}$

5. Let $f : (3, 6) \rightarrow (4, 7)$ be a function defined by $f(x) = x + \left[\frac{x}{3} \right]$, then $f^{-1}(x)$ is equal to

(where $[.]$ denotes greatest integer function)

(A) $x - \left[\frac{x}{4} \right]$

(B) $x + 1$

(C) $x + \left[\frac{x}{3} \right]$

(D) $2x - 1$

6. Consider the following statements

$S_1 : \lim_{x \rightarrow \infty} \frac{\sin(5^x)}{5^x} = 1$

$S_2 : \lim_{n \rightarrow \infty} \frac{(n+2)! + (n+3)!}{(n+4)!} = 0$

$S_3 : \lim_{x \rightarrow \infty} \left(\frac{1}{x^2} + \frac{2}{x^2} + \dots + \frac{x}{x^2} \right) = \frac{1}{2}, \text{ (where } x \in \mathbb{N} \text{)}$

$S_4 : \lim_{x \rightarrow 0} \frac{\sin x^\circ}{x} = 1$

State in order, whether S_1, S_2, S_3, S_4 are true or false

(A) TTFT

(B) FTTT

(C) TTTT

(D) FTTF

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Which of the following is/are correct

- (A) Fundamental period of $f(x) = \cos(\sin x) + \cos(\cos x)$ is π
 (B) Range of the function $f(x) = \cos(2\sin x)$ is $[\cos 2, 1]$
 (C) $f(x) = 0$ is a periodic function

(D) $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \mathbb{R}, f(x) = \sin x$ is a many-one function

8. Which of the following function is/are bijective functions

- (A) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3 + 2x^2 - x + 1$ (B) $f : [0, \infty) \rightarrow [1, \infty), f(x) = \sqrt{1+x^2}$
 (C) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3 + x + 1$ (D) $f : [-1, 1] \rightarrow [0, \sin 1], f(x) = \sin \sqrt{1-x^2}$

Space for Rough Work

9. If $x \in (0, 1)$ then $\tan^{-1} \frac{\sqrt{1-x^2}}{1+x}$ is equal to

- (A) $\frac{\cos^{-1} x}{2}$ (B) $\cos^{-1} \sqrt{\frac{1+x}{2}}$ (C) $\sin^{-1} \sqrt{\frac{1-x}{2}}$ (D) $\frac{\tan^{-1} x}{2}$

10. Let $k \in \mathbb{N}$ and $a \in \mathbb{R}^+$ ($a \neq 1$) then $\lim_{n \rightarrow \infty} n^k \left(a^{\frac{1}{n}} - 1 \right) \left(\sqrt{\frac{n-1}{n}} - \sqrt{\frac{n+1}{n+2}} \right)$ is

- (A) 0 if $k \in \{1, 2\}$ (B) $-\ln a$ if $k = 3$
(C) Does not exist if $k \geq 4$ and $a \in (0, 1)$ (D) Does not exist if $k \geq 4$ and $a > 1$

11. If $\lim_{x \rightarrow \infty} \left(((x-1)(x-2)(x-3)(x-4))^{1/4} - x \right) = \lambda$ then $|\lambda|$ equals

- (A) $\frac{5}{2}$ (B) 5 (C) $\lim_{x \rightarrow 0} \frac{\sin 5x}{2x}$ (D) $\lim_{x \rightarrow 0} \frac{\tan^{-1} 5x}{x}$

12. If $x \in (0, 1)$ and $f(x) = \sec \left\{ \tan^{-1} \left(\frac{\sin(\cos^{-1} x) + \cos(\sin^{-1} x)}{\cos(\cos^{-1} x) + \sin(\sin^{-1} x)} \right) \right\}$ then $\sum_{r=2}^5 f\left(\frac{1}{r}\right)$ equals

- (A) $\lim_{x \rightarrow 0} \frac{e^{14x} - 1}{x}$ (B) $\lim_{x \rightarrow 0} \frac{\log_e(1+12x)}{x}$ (C) $\lim_{x \rightarrow 0} \frac{\sin 12x}{x}$ (D) $\lim_{x \rightarrow 0} \frac{\tan^{-1} 11x}{x}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Let $\cos^{-1} \sqrt{x^3 - x + 1} + \cos^{-1} \sqrt{x - x^3} + \cos^{-1} \sqrt{1 - |y|} = \frac{2\pi}{3}$ then the value of $2|y|$ equals
- Let $f(x) = \tan^{-1}x$ and $g(x) = \tan \sqrt{x}$. If $S = \sum_{r=1}^5 \left[f(g(r^4)) \right]$ (where $[.]$ denotes greatest integer

function) then value of $\frac{|S|}{3}$ equals

Space for Rough Work

3. Let $S = \cot^{-1} \frac{25}{2} + \cot^{-1} \frac{36}{2} + \cot^{-1} \frac{49}{2} + \dots$ upto n-terms and $\tan S = f(n)$. If $\lim_{n \rightarrow \infty} f(n) = \frac{p}{q}$ (where p & q are co-prime natural numbers) then the value of $\frac{|p-q|}{4}$
4. Number of solutions of the equation $\tan^{-1}(e^{-x}) + \cot^{-1}(|\log_e x|) = \frac{\pi}{2}$ is
5. If $\lim_{x \rightarrow 0} \frac{a + b \sin x - \cos x + ce^x}{x^3}$ exists finitely (where a, b, c $\in \mathbb{R}$) then $\frac{|a+b+4c|}{4}$ equals
6. Let $S_n = \frac{2.6}{2} + \frac{6.12}{3} + \frac{12.20}{4} + \frac{20.30}{5} + \dots$ upto n-terms. Find the value of $\frac{S_{20}}{105}$

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	18
I(ii)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator